Network Administration and Security Project Report

EVE-NG Platform Implementation

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1 Project Introduction

This project, developed by **Idbouzghiba Houssam** and **Assioui Abdelilah**, focuses on **network administration and security** using the **EVE-NG** platform and virtual machines. It covers a wide range of technologies, including the configuration of **Palo Alto** firewalls (zones, policies, and dynamic routing), **IP addressing** and profiles, Layer 2 technologies such as **EtherChannel**, **VTP**, and **standby**, **inter-VLAN routing** using multilayer switches, and the implementation of **Active Directory** for centralized user and resource management. The project also addresses **redundancy** at both Layer 2 and Layer 3, **OSPF implementation** on routers, firewalls, and Layer 3 switches, **VPN setup** between two sites, and connectivity to the **internet** via **lab-host bridging** to the home network with **PAT/NAT overload**. All these technologies were successfully implemented and tested within our **lab environment**. This report provides a detailed explanation of each technology and describes how it was applied in our topology.

2 Acknowledgment

We would like to express our sincere gratitude to our teacher and mentor, **Madam Kissi Chaimae**, for her invaluable guidance throughout the semester. She consistently answered our questions, assisted us with various challenges, and provided instructions that taught us how to learn new concepts independently. While this project represents only a small fraction of the knowledge she imparted, we hope it offers at least a glimpse of her tremendous impact on our learning journey.

3 Software Used

3.1 VMware

VMware was used as a hypervisor to host our **EVE-NG** virtual machine. It was installed directly from the official VMware website. This allowed us to run multiple virtual devices on a single physical machine, providing a stable environment for our lab topology and ensuring smooth performance during network simulations.

3.2 EVE-NG

EVE-NG was the primary software used to create and manage our network topology. The software is installed by downloading the ISO image, configuring it in **VMware**, and assigning a static IP address within the home network range. The web-based GUI can then be accessed by entering this IP in a browser. For collaborative work, we configured **port forwarding** on the home router to forward requests to the private IP `192.168.11.120`, allowing my partner Houssam to access the GUI remotely using the public IP.

3.3 WinSCP

WinSCP was used to securely transfer software images (such as **Windows, Ubuntu, and Palo Alto**) from the local computer to the appropriate **EVE-NG folders**. Trans-

fers were performed over FTP using the EVE-NG machine's IP `192.168.20.120`. This tool simplified the upload of virtual machine images and ensured proper organization within the lab environment.

3.4 SecureCRT

SecureCRT was used to access the terminals of VMs inside the **EVE-NG** lab, including routers, switches, and VPCs. It provided a reliable **SSH/Telnet interface**, allowing us to configure devices efficiently and monitor their behavior during testing.

3.5 Client Pack

The **EVE-NG Client Pack** was installed from the official website and includes tools such as **Wireshark** for network traffic capture and **UltraVNC Viewer** to access the GUI of Windows and Ubuntu machines. While many tools were included in the client pack, only a few were actively used for monitoring and configuration during this project.

3.6 VMs inside EVE-NG

A variety of **virtual machines (VMs)** were deployed within **EVE-NG** using nested virtualization. Images were uploaded to the correct folders via **WinSCP**. The VMs included **Cisco routers and switches**, **Windows 10** and **7**, Windows **Server** 2016, **Ubuntu Server**, and **Palo Alto** firewalls, enabling a realistic, fully functional lab environment for testing network configurations and security setups

4 Topology

4.1 SiteA

4.1.1 End Hosts

SiteA has 3 VLANs in the LAN zone, each containing 3 active end hosts. The end hosts include client machines using DHCP to obtain their IP addresses and server machines with static IPs. Additionally, a Windows 10 machine is deployed in the **MGMT zone** to manage the Palo Alto firewall, and an Ubuntu server is located in the DMZ zone, functioning as a web server accessible through controlled firewall policies.

4.1.2 Access Switches

Three access switches serve the LAN, each connected to 3 end hosts via downlink access ports corresponding to the VLANs. Each access switch also has 2 uplink trunk ports connected to the Layer 3 switches, enabling **VLAN tagging** and inter-VLAN traffic forwarding. The design was set up to ensure proper segmentation of VLAN traffic and **connectivity** to higher-level switches.

4.1.3 Layer 3 Switches

Two Layer 3 switches are deployed with 2 inter-switch links configured as routed ports bundled into an EtherChannel for redundancy and increased bandwidth. Each Layer 3

switch has 3 downlink trunk ports, each connecting to one of the access switches, allowing **inter-VLAN routing** and communication between all VLANs in **SiteA**.

4.1.4 Palo Alto Firewall – PaloAlto-A

A **Palo Alto firewall** is deployed at the ingress of SiteA. It is managed by a Windows 10 machine through the management interface and connects to the two **Layer 3 switches** in the LAN zone. It also connects to an access switch in the **DMZ zone** and is linked to two ISP routers with distinct IP subnets to provide internet connectivity and redundancy. The firewall enforces **security policies** and manages traffic between the internal LAN, DMZ, and WAN.

4.2 WAN Part

The WAN consists of **four ISP routers** configured with OSPF for dynamic routing. These routers are interconnected via point-to-point links to ensure **network redundancy**. The top two ISP routers connect to an EDGE-ISP router, which bridges traffic to a network node linking the lab to the home network. This allows external connectivity for testing internet access from lab hosts.

4.3 SiteB

SiteB contains 3 end hosts, including a VPC for quick ping tests, a Windows 7 machine, and an Ubuntu server, all connected to a single access switch in the default **VLAN1**. The access switch uplink connects to the **Palo Alto firewall – PaloAlto-B**, which is also managed via a Windows 10 machine through its management interface. The firewall connects to two ISP routers to provide internet connectivity and redundancy, enabling **SiteB** to communicate with SiteA and the WAN.

5 IP Addressing

5.1 SiteA

5.1.1 VLAN10 (Sales)

IP range - 192.168.20.0/27
Default gateway - 192.168.20.30

5.1.2 VLAN20 (HR)

IP range - 192.168.20.32/27
Default gateway - 192.168.20.50

5.1.3 VLAN30 (Control)

IP range - 192.168.20.64/27
Default gateway - 192.168.20.80
Reserved address - 192.168.20.66 for AD domain controller

5.1.4 EtherChannel link

IP range - 192.168.20.96/30
SW1 - 192.168.20.97
SW2 - 192.168.20.98

5.1.5 Point-to-Point 1

IP range - 192.168.20.100/30
SW1 (G1/0) - 192.168.20.101
PaloAlto-A (eth1/1) - 192.168.20.102

5.1.6 Point-to-Point 2

IP range - 192.168.20.104/30
SW2 (G1/0) - 192.168.20.105
PaloAlto-A (eth1/2) - 192.168.20.106

5.1.7 MGMT link

IP range - 10.10.10.0/30
Admin-Mgmt (Win10) - 10.10.10.1
PaloAlto-A (mgmt) - 10.10.10.2

5.1.8 DMZ / Web Server

IP range - 172.16.20.0/24
Web_Server (Ubuntu) - 172.16.20.1
PaloAlto-A (eth1/3) - 172.16.20.254

5.1.9 SVIs on SW1

SVI for VLAN20 - 192.168.20.62
SVI for VLAN10 - 192.168.20.30
SVI for VLAN30 - 192.168.20.94

5.1.10 SVIs on SW2

SVI for VLAN20 - 192.168.20.61
SVI for VLAN10 - 192.168.20.29
SVI VLAN30 - 192.168.20.93

5.1.11 Standby IP on SW1 & SW2

for VLAN10 - 192.168.20.30
for VLAN20 - 192.168.20.50
for VLAN30 - 192.168.20.80

5.2 WAN

5.2.1 Point-to-Point 1 (Firewall - ISP1)

IP range - 10.0.1.0/30
PaloAlto-A (eth1/4) - 10.0.1.1
ISP1 (fa0/0) - 10.0.1.2

5.2.2 Point-to-Point 2 (Firewall - ISP2)

IP range - 10.0.2.0/30
PaloAlto-A (eth1/5) - 10.0.2.1
ISP2 (fa0/0) - 10.0.2.2

5.2.3 Point-to-Point 3 (ISP3 - Firewall)

IP range - 10.0.3.0/30
PaloAlto-B (eth1/1) - 10.0.3.1
ISP3 (fa0/0) - 10.0.3.2

5.2.4 Point-to-Point 4 (ISP4 - Firewall)

IP range - 10.0.4.0/30
PaloAlto-B (eth1/2) - 10.0.4.1
ISP4 (fa0/0) - 10.0.4.2

5.2.5 Point-to-Point 5 (ISP1 - ISP2)

IP range - 10.0.5.0/30
ISP1 (f2/0) - 10.0.5.1
ISP2 (f1/0) - 10.0.5.2

5.2.6 Point-to-Point 6 (ISP1 - ISP3)

IP range - 10.0.6.0/30
ISP1 (f3/0) - 10.0.6.1
ISP3 (f3/0) - 10.0.6.2

5.2.7 Point-to-Point 7 (ISP2 - ISP4)

IP range - 10.0.7.0/30
ISP2 (f2/0) - 10.0.7.1
ISP4 (f2/0) - 10.0.7.2

5.2.8 Point-to-Point 8 (ISP3 - ISP4)

IP range - 10.0.8.0/30
ISP3 (f1/0) - 10.0.8.1
ISP4 (f2/0) - 10.0.8.2

5.2.9 Point-to-Point 9 (ISP1 - Edge-ISP)

IP range - 10.0.9.0/30
ISP1 (f1/0) - 10.0.9.1
Edge-ISP (f0/0) - 10.0.9.2

5.2.10 Point-to-Point 10 (ISP3 - Edge-ISP)

IP range - 10.0.10.0/30
ISP3 (f1/0) - 10.0.10.1
Edge-ISP (f1/0) - 10.0.10.2

5.2.11 Home

Edge-ISP (f2/0) - DHCP from home router
Network IP range - 192.168.11.0/24

5.3 SiteB

5.3.1 LAN zone

IP range - 192.168.20.128/27
PaloAlto-B (eth1/3) - 192.168.20.157
Remaining hosts - DHCP

5.3.2 MGMT zone

IP range - 10.10.20.0/30
Admin-Mgmt2 (Win) - 10.10.20.1
PaloAlto-B (mgmt) - 10.10.20.2

6 Technologies Implemented

6.1 Subnetting

6.1.1 General explanation

Subnetting is the process of dividing a larger IP network into smaller, more manageable sub-networks. It improves network organization, reduces broadcast traffic, enhances security, and allows more efficient use of IP address space. By assigning different subnet masks, network administrators can separate departments, point-to-point links, and infrastructure segments while maintaining logical and scalable addressing.

6.1.2 Usage in the lab

In Site A, the main IP prefix `192.168.20.0/24` was variably subnetted to match the needs of different VLANs and infrastructure links. Subnets `192.168.20.0/27`, `192.168.20.32/27`, and `192.168.20.64/27` were assigned to VLAN10 (Sales), VLAN20 (HR), and VLAN30 (Control) respectively. Smaller `/30` subnets (`192.168.20.96/30`, `192.168.20.100/30`,

and `192.168.20.104/30`) were used for point-to-point links between SW1–SW2 (EtherChannel), SW1–PaloAltoA, and SW2–PaloAltoB, ensuring efficient address utilization and clear separation of traffic.

6.2 Active Directory

6.2.1 General

Active Directory (**AD**) is a directory service by Microsoft that provides centralized management of users, computers, and network resources. It allows domain-based authentication, policy enforcement, and DNS integration for easier administration and security.

6.2.2 Implementation in the lab

A **Domain Controller** was set up on the Windows Server 2016 machine in VLAN30 (Control) with static IP 192.168.20.66, reserved in the DHCP pool of the multilayer switches. Windows 10 end hosts HR1, HR3, Sales1, and Sales3 joined the domain, each with a unique **user account**: Houssam, Youssef, Anwar, Amin. Passwords were set to never expire to simplify demonstration purposes. Another Windows 10 machine in VLAN30 was used to access the admin account of the DC. The **Domain Controller IP** was assigned as the primary DNS server via DHCP and excluded from dynamic allocation on both L3 switches.

6.3 VLANs

6.3.1 General

VLANs (Virtual LANs) logically segment a network at Layer 2, isolating traffic between different groups while sharing the same physical infrastructure. This improves **security**, performance, and manageability.

6.3.2 Implementation in the lab

Three VLANs were implemented in **SiteA**: VLAN10 (Sales), VLAN20 (HR), and VLAN30 (Control). VLANs were initially created on SW1 and automatically propagated to all other switches via the VTP domain `inside-lan`.

6.4 Inter-VLAN Routing

6.4.1 General

Inter-VLAN routing allows communication between devices in different VLANs using a Layer 3 device such as a multilayer switch or router. This is typically done via **SVIs (Switched Virtual Interfaces)**, each acting as a default gateway for a VLAN.

6.4.2 Implementation in the lab

On both L3 switches (SW1 and SW2) in SiteA, three SVIs were configured, each with an IP acting as the default gateway for end hosts in each VLAN. HSRP/standby was later configured on a logical IP address for each VLAN. The SVIs take turns using this logical IP, providing **redundancy** and failover for end hosts.

6.5 Standby

6.5.1 General

Standby/HSRP (Hot Standby Router Protocol) provides a **logical default gateway IP** shared by multiple routers or switches. If one device fails, the backup device automatically assumes the IP, ensuring continuous connectivity for hosts.

6.5.2 Implementation in the lab

On SW1 and SW2, standby was configured over IPs 192.168.20.30, 192.168.20.50, 192.168.20.80 for VLANs 10, 20, 30. Each SVI shares a logical IP address, so if SW1 fails, SW2 SVIs take over automatically. End hosts continue using the same default gateway IP without interruption.

6.6 EtherChannel

6.6.1 General

EtherChannel bundles multiple physical links into a single logical link to increase **bandwidth** and provide link redundancy. If one physical interface fails, traffic continues over the remaining links.

6.6.2 Implementation in the lab

EtherChannel was configured on the link between SW1 and SW2 with 2 physical links. The logical IPs 192.168.20.97 (SW1) and 192.168.20.98 (SW2) were assigned to the EtherChannel interfaces. This provides two paths between the switches, maintaining connectivity if a single link fails.

6.7 OSPF

6.7.1 General

OSPF (Open Shortest Path First) is a **dynamic routing protocol** that enables routers to exchange routing information and calculate the shortest path to networks within an autonomous system. Passive interfaces can be used to prevent unnecessary routing messages. OSPF was basically configured on the same area, area 0, on all devices in the lab for simplicity.

6.7.2 Implementation in the lab

OSPF was configured on:

- Uplink interfaces of SW1 and SW2 (G1/0) - All three SVIs on both L3 switches - All interfaces of PaloAlto-A - All interfaces of the ISP routers, including Edge-ISP - All interfaces of PaloAlto-B

Interfaces connected to end hosts on both firewalls were set to passive to prevent OSPF messages from reaching end devices, while still advertising the networks. This setup allows all routers and firewalls to dynamically learn LANs, VLANs, and point-to-point links on **SiteA** and reach SiteB networks.

6.8 PAT to bridge to home network

6.8.1 General

Bridging a virtual lab to a physical network allows simulated devices to access real external networks. NAT with PAT translates **private IP addresses** to a public or shared IP, enabling **internet access** for internal hosts.

6.8.2 Usage in the lab

A management network node (`mgmt(cloud0)`) was connected to the EDGE-ISP router to bridge the EVE-NG lab to the home network. PAT overload was configured on interface `fa2/0` of the EDGE-ISP router, which receives its IP address via DHCP from the home router. This allowed all internal hosts to successfully reach **internet IP addresses** such as `8.8.8.8`.

6.9 VPN (Site-to-Site using Palo Alto)

A **site-to-site VPN** was configured between PaloAltoA and PaloAltoB using tunnel interface `tunnel.1`. IKE gateways and IKE profiles were created on both firewalls, and security policies were defined to allow traffic from the LAN zone to the VPN zone. This configuration enables **secure communication** between both sites.

The IPsec Site-to-Site VPN is configured using IKEv2 for Phase 1, with **AES-256 encryption**, SHA-256 authentication, and Diffie-Hellman Group 14 (2048-bit). The Phase 1 lifetime is set to 8 hours. In IKE Phase 2 (IPsec), AES-256 encryption and SHA-256 authentication are also used, with Perfect Forward Secrecy (PFS) enabled using Group 14, and a lifetime of 1 hour. The tunnel is configured on interfaces on both firewalls, assigned to the VPN zone, and uses **proxy IDs** to map Site A subnets to Site B subnets.

6.10 Default Route

6.10.1 General

A **default route** directs traffic destined for unknown networks toward a specified next hop, typically an ISP router, enabling access to **external networks**.

6.10.2 Usage in the lab

A default route was configured on the EDGE-ISP router with next hop `192.168.11.1`, corresponding to the home router. This default route was advertised through OSPF to all routers, then propagated to firewalls and multilayer switches, allowing **all internal hosts** to reach the internet.

6.11 VTP (VLAN Trunking Protocol)

6.11.1 General explanation

VTP is a **Cisco protocol** that allows centralized VLAN management across multiple switches. VLAN information is distributed using VTP advertisements within a VTP

domain, reducing configuration errors and ensuring consistency. Switches synchronize VLAN databases based on the **VTP revision number**.

6.11.2 Usage in the lab

A VTP domain named `inside-lan` was created on **SW1**, which acted as the primary VTP server. After configuring the three VLANs on SW1, other switches in Site A automatically joined the domain upon receiving VTP advertisements with a higher revision number. As a result, all switches dynamically learned and created the VLANs without manual configuration on each device.

6.12 DHCP

6.12.1 General

DHCP (Dynamic Host Configuration Protocol) automates IP address assignment to end hosts, reducing manual configuration errors and simplifying network management. It dynamically provides essential network parameters such as the IP address, subnet mask, default gateway, and DNS server. In redundant network designs, DHCP must be carefully coordinated with gateway redundancy mechanisms to ensure consistent and reliable address allocation.

6.12.2 Usage in the lab

DHCP services were configured on both **SW1** and **SW2**, acting as Layer 3 switches, to provide dynamic IP addressing for all end hosts in Site A. Three DHCP pools were created, corresponding to the three VLANs: 192.168.20.0/27 (VLAN10 – Sales), 192.168.20.32/27 (VLAN20 – HR), and 192.168.20.64/27 (VLAN30 – Control).

The IP address 192.168.20.66 was **statically reserved** for the Active Directory Domain Controller, ensuring a fixed address for critical services. Several addresses were explicitly excluded from dynamic allocation, including the standby (HSRP) virtual IP addresses (192.168.20.30, 192.168.20.50, 192.168.20.80) and the SVI addresses assigned to SW1 and SW2 (192.168.20.29, 192.168.20.61, 192.168.20.62, 192.168.20.93, 192.168.20.94).

This configuration guarantees that DHCP-assigned addresses do not conflict with infrastructure IPs while maintaining **redundancy**, consistency, and high availability across the network.

6.13 Firewalling

6.13.1 Management

General
Firewall management refers to the secure administration of firewall devices through dedicated interfaces, allowing configuration, monitoring, and troubleshooting without exposing control access to production traffic. Palo Alto firewalls provide both CLI and web-based management interfaces.
Usage in the lab

After configuring the management (mgmt) interface IP address of each firewall (PaloAlto-A and PaloAlto-B) via the CLI, both devices were managed through the web interface (HTTPS).
Access was performed from a Windows 10 machine located in a dedicated management segment, using the private IP address assigned to the mgmt interface.
Initial access was achieved using the default credentials (admin / admin), after which configuration was performed entirely through the graphical interface.

6.13.2 Interfaces

General
Firewall interfaces define the connection points between different network segments. Each interface is assigned an IP address, mapped to a security zone, and associated with a network profile that controls allowed management services such as ping or HTTPS.
Usage in the lab
All firewall interfaces were assigned IP addresses corresponding to their respective network segments. A management profile allowing ICMP (ping) was applied to enable reachability testing. Each interface was placed in an appropriate security zone.
For Site A, interfaces eth1/1 and eth1/2 were assigned to the LAN zone, eth1/3 to the DMZ zone, and eth1/4 to the WAN zone.
For Site B, interfaces eth1/1 and eth1/2 were assigned to the WAN zone, while eth1/3 was assigned to the LAN zone.
Additionally, a tunnel interface (`tunnel.1`) was created on both firewalls to support the IPsec site-to-site VPN between Site A and Site B.

6.13.3 Virtual Router (Default Router)

General
A virtual router in Palo Alto firewalls is a logical routing instance responsible for packet forwarding, dynamic routing protocols, and static route management. Interfaces must be associated with a virtual router to participate in routing.
Usage in the lab
The default virtual router was used as the main routing instance on both firewalls. All physical and tunnel interfaces were associated with this router. OSPF was configured within the virtual router to advertise connected networks and dynamically learn routes. Two static routes were defined:

- A default route (0.0.0.0/0) pointing to the directly connected ISP router as the next hop
- A VPN route using `tunnel.1` as the next hop to reach the remote LAN networks (192.168.20.0/24 and 192.168.20.128/27) between Site A and Site B

6.13.4 Zones

General
Security zones are a core concept in Palo Alto firewalls, grouping interfaces with similar trust levels. Traffic is controlled based on source and destination zones rather than physical interfaces.
Usage in the lab

On PaloAlto-A, three zones were created: LAN, WAN, and DMZ.
On PaloAlto-B, two zones were created: LAN and WAN.
The LAN zone represents the trusted internal network, while the WAN zone represents untrusted external networks.
A dedicated VPN zone was also created on both firewalls, with the tunnel interface (`tunnel.1`) assigned to it to logically separate encrypted VPN traffic.

6.13.5 Security Policies

General
Security policies define which traffic is permitted or denied between zones. Policies are evaluated based on source zone, destination zone, application, and service.
Usage in the lab
On PaloAlto-A, security policies were configured to allow traffic from LAN to WAN and from WAN to DMZ.
On PaloAlto-B, policies were configured to allow traffic from LAN to WAN.
Additional VPN security policies were created on both firewalls to allow traffic between the LAN zone and the VPN zone, enabling secure inter-site communication over the IPsec tunnel.